

Name: _____

Date: _____

SURDS, (RATIONALISING THE DENOMINATOR)

LO: I can rationalise the denominator of a surd fraction.

Rationalising

Rationalising means removing the surd from the **denominator** (bottom) of a fraction by multiplying top and bottom by an appropriate expression.

Methods: For $\frac{a}{b}$, multiply by $\frac{b}{b}$. For $\frac{a}{c+d}$, multiply by $\frac{c-d}{c-d}$ (conjugate).

Quick examples

●	$\frac{1}{2} = \frac{2}{2}$	multiply by $\frac{2}{2}$
●	$\frac{6}{3} = \frac{6}{3} = 2$	simplify after
●	$\frac{1}{2+3}$ uses conjugate $2-3$	difference of squares
●	$\frac{1}{5} = \frac{1}{5}$	must multiply properly
●	$\frac{3}{2} = \frac{3}{2}$	$= 3 \frac{2}{2}$

Key idea

Multiply numerator and denominator by the surd (or conjugate) to eliminate the irrational denominator.

$$\frac{a}{b} \times \frac{b}{b} = \frac{a}{b/b}$$

The denominator becomes rational

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple rationalising

Rationalise $\frac{5}{3}$.

$$\begin{aligned} &\text{Multiply by } \frac{3}{3} \\ &= \frac{5}{3} \times \frac{3}{3} \\ &= \frac{5}{1} \end{aligned}$$

$$\frac{5}{3}$$

Multiply top and bottom by 3.

Example 2 – simplify after

Rationalise $\frac{6}{2}$.

$$\begin{aligned} &\text{Multiply by } \frac{2}{2}: \frac{6}{2} = \frac{6}{2} \\ &\text{Simplify: } \frac{6}{2} = 3 \\ &= 3 \end{aligned}$$

$$3$$

Always simplify the fraction after rationalising.

Example 3 – conjugate method

Rationalise $\frac{1}{3+2}$.

$$\begin{aligned} &\text{Multiply by } \frac{3-2}{3-2} \\ &\text{Numerator: } 3-2 \\ &\text{Denominator: } 9-2 = 7 \end{aligned}$$

$$\frac{3-2}{7}$$

Use the conjugate: $(a+b)(a-b) = a^2 - b^2$.

Example 4 – harder conjugate

Rationalise $\frac{4}{5-1}$.

$$\begin{aligned} &\text{Multiply by } \frac{5+1}{5+1} \\ &\text{Num: } 4(5+1) = 4 \times 5 + 4 \\ &\text{Den: } 5-1 = 4 \end{aligned}$$

$$5+1$$

Simplify the fraction: $\frac{4 \times 5 + 4}{4} = 5 + 1$.

YOUR TURN – PRACTICE (deliberate practice)

1. Simple rationalising

Rationalise the denominator.

- a) $\frac{1}{5}$
- b) $\frac{3}{7}$
- c) $\frac{4}{2}$
- d) $\frac{10}{5}$

2. Simplify after rationalising

Rationalise and fully simplify.

- a) $\frac{6}{3}$
- b) $\frac{8}{2}$
- c) $\frac{12}{6}$
- d) $\frac{15}{5}$

3. Conjugate method (addition)

Rationalise using the conjugate.

- a) $\frac{1}{1+2}$
- b) $\frac{1}{3+5}$
- c) $\frac{2}{4+3}$
- d) $\frac{5}{2+7}$

4. Conjugate method (subtraction)

Rationalise the denominator.

- a) $\frac{1}{3-1}$
- b) $\frac{3}{5-2}$
- c) $\frac{4}{7-3}$
- d) $\frac{6}{5-2}$

5. Mixed rationalising

Rationalise and simplify fully.

- a) $\frac{2}{8}$
- b) $\frac{3}{12}$
- c) $\frac{1}{1+3}$
- d) $\frac{5+1}{5-1}$
- e) $\frac{6}{3-2}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To rationalise $\frac{a}{b}$, multiply top and bottom by b ☐

Correct answer: _____
Reason: _____

b) $\frac{1}{2} = \frac{1}{2}$ ☐

Correct answer: _____
Reason: _____

c) The conjugate of $(a + b)$ is $(a - b)$ ☐

Correct answer: _____
Reason: _____

d) $\frac{4}{2} = 4 \div 2$ ☐

Correct answer: _____
Reason: _____

e) $(a + b)(a - b) = a^2 - b$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Show that $\frac{1}{2+1} + \frac{1}{2-1} = 2 \div 2$ by rationalising each fraction.

Expression: _____

Simplified: _____

b) Simplify $3 + \frac{1}{3-1} - 3 - \frac{1}{3+1}$. Give your answer in the form $a \div b$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

